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CS3502 | Section W03
Assignment 2 Deliverables
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Test 1

```
cconnell@cconnell:~/cs3502/assignment2/c-programs$ ./producer 1 5 & ./consumer 1 5
[1] 5772
Producer 1: Starting to produce 5 items
Producer 1: Produced value 1000
Consumer 1: Starting to consume 5 items
Consumer 1: Consumed value 1000 from Producer 1
Producer 1: Produced value 1001
Consumer 1: Consumed value 1001 from Producer 1
Producer 1: Produced value 1002
Consumer 1: Consumed value 1002 from Producer 1
Producer 1: Produced value 1003
Consumer 1: Consumed value 1003 from Producer 1
Producer 1: Produced value 1004
Consumer 1: Consumed value 1004 from Producer 1
Producer 1: Finished producing 5 items
Consumer 1: Finished consuming 5 items
[1]+  Done                    ./producer 1 5
cconnell@cconnell:~/cs3502/assignment2/c-programs$
```

Test 2

```
cconnell@cconnell:~/cs3502/assignment2/c-programs$ ./producer 1 10 & ./producer 2 10 & ./producer 3 10 & ./consumer 1 30
[1] 5823
[2] 5824
[3] 5825
Producer 1: Starting to produce 10 items
Producer 2: Starting to produce 10 items
Producer 2: Produced value 2000
Producer 1: Produced value 1000
Producer 3: Starting to produce 10 items
Producer 3: Produced value 3000
Consumer 1: Starting to consume 30 items
Consumer 1: Consumed value 2000 from Producer 2
Producer 3: Produced value 3001
Producer 1: Produced value 1001
Consumer 1: Consumed value 1000 from Producer 1
Producer 3: Produced value 3002
Producer 2: Produced value 2001
Consumer 1: Consumed value 3000 from Producer 3
Producer 1: Produced value 1002
Producer 1: Produced value 1003
Producer 3: Produced value 3003
Producer 2: Produced value 2002
Producer 1: Produced value 1004
Consumer 1: Consumed value 3001 from Producer 3
Producer 3: Produced value 3004
Producer 1: Produced value 1005
Consumer 1: Consumed value 1001 from Producer 1
Producer 2: Produced value 2003
Consumer 1: Consumed value 3002 from Producer 3
Producer 2: Produced value 2004
Consumer 1: Consumed value 2001 from Producer 2
Producer 3: Produced value 3005
Consumer 1: Consumed value 1002 from Producer 1
Producer 1: Produced value 1006
Consumer 1: Consumed value 1003 from Producer 1
Producer 3: Produced value 3006
Consumer 1: Consumed value 3003 from Producer 3
Producer 2: Produced value 2005
Consumer 1: Consumed value 2002 from Producer 2
Producer 1: Produced value 1007
Consumer 1: Consumed value 1004 from Producer 1
Producer 2: Produced value 2006
Consumer 1: Consumed value 3004 from Producer 3
Producer 3: Produced value 3007
Consumer 1: Consumed value 1005 from Producer 1
Producer 1: Produced value 1008
Consumer 1: Consumed value 2003 from Producer 2
Producer 3: Produced value 3008
Consumer 1: Consumed value 2004 from Producer 2
Producer 2: Produced value 2007
Consumer 1: Consumed value 3005 from Producer 3
Producer 1: Produced value 1009
Producer 1: Finished producing 10 items
Consumer 1: Consumed value 1006 from Producer 1
Producer 3: Produced value 3009
Producer 3: Finished producing 10 items
Consumer 1: Consumed value 3006 from Producer 3
Producer 2: Produced value 2008
Consumer 1: Consumed value 2005 from Producer 2
Consumer 1: Consumed value 1007 from Producer 1
Producer 2: Produced value 2009
Consumer 1: Consumed value 2006 from Producer 2
Producer 2: Finished producing 10 items
Consumer 1: Consumed value 3007 from Producer 3
Consumer 1: Consumed value 1008 from Producer 1
Consumer 1: Consumed value 3008 from Producer 3
Consumer 1: Consumed value 2007 from Producer 2
Consumer 1: Consumed value 1009 from Producer 1
Consumer 1: Consumed value 3009 from Producer 3
Consumer 1: Consumed value 2008 from Producer 2
Consumer 1: Consumed value 2009 from Producer 2
Consumer 1: Finished consuming 30 items
[1]   Done                    ./producer 1 10
[2]-  Done                    ./producer 2 10
[3]+  Done                    ./producer 3 10
cconnell@cconnell:~/cs3502/assignment2/c-programs$
```

Test 3

```
cconnell@cconnell:~/cs3502/assignment2/c-programs$ ./producer 1 20 & ./consumer 1 10 & ./consumer 2 10
[1] 5805
[2] 5806
Consumer 1: Starting to consume 10 items
Producer 1: Starting to produce 20 items
Producer 1: Produced value 1000
Consumer 1: Consumed value 1000 from Producer 1
Consumer 2: Starting to consume 10 items
Producer 1: Produced value 1001
Consumer 2: Consumed value 1001 from Producer 1
Producer 1: Produced value 1002
Consumer 1: Consumed value 1002 from Producer 1
Producer 1: Produced value 1003
Consumer 1: Consumed value 1003 from Producer 1
Producer 1: Produced value 1004
Consumer 2: Consumed value 1004 from Producer 1
Producer 1: Produced value 1005
Consumer 1: Consumed value 1005 from Producer 1
Producer 1: Produced value 1006
Consumer 2: Consumed value 1006 from Producer 1
Producer 1: Produced value 1007
Consumer 1: Consumed value 1007 from Producer 1
Producer 1: Produced value 1008
Consumer 2: Consumed value 1008 from Producer 1
Producer 1: Produced value 1009
Consumer 1: Consumed value 1009 from Producer 1
Producer 1: Produced value 1010
Consumer 2: Consumed value 1010 from Producer 1
Producer 1: Produced value 1011
Consumer 1: Consumed value 1011 from Producer 1
Producer 1: Produced value 1012
Consumer 2: Consumed value 1012 from Producer 1
Producer 1: Produced value 1013
Consumer 1: Consumed value 1013 from Producer 1
Producer 1: Produced value 1014
Consumer 2: Consumed value 1014 from Producer 1
Producer 1: Produced value 1015
Consumer 2: Consumed value 1015 from Producer 1
Producer 1: Produced value 1016
Consumer 1: Consumed value 1016 from Producer 1
Producer 1: Produced value 1017
Consumer 2: Consumed value 1017 from Producer 1
Producer 1: Produced value 1018
Consumer 1: Consumed value 1018 from Producer 1
Producer 1: Produced value 1019
Consumer 2: Consumed value 1019 from Producer 1
Producer 1: Finished producing 20 items
Consumer 1: Finished consuming 10 items
Consumer 2: Finished consuming 10 items
[1]-  Done                    ./producer 1 20
[2]+  Done                    ./consumer 1 10
cconnell@cconnell:~/cs3502/assignment2/c-programs$
```

Test 4

```
cconnell@cconnell:~/cs3502/assignment2/c-programs$ ./producer 1 15 & ./producer 2 15 & ./consumer 1 15 & ./consumer 2 15
[1] 5784
[2] 5785
[3] 5786
Producer 2: Starting to produce 15 items
Producer 2: Produced value 2000
Consumer 1: Starting to consume 15 items
Producer 1: Starting to produce 15 items
Producer 1: Produced value 1000
Consumer 2: Starting to consume 15 items
Consumer 2: Consumed value 2000 from Producer 2
Consumer 1: Consumed value 1000 from Producer 1
Producer 2: Produced value 2001
Consumer 2: Consumed value 2001 from Producer 2
Producer 1: Produced value 1001
Consumer 1: Consumed value 1001 from Producer 1
Producer 2: Produced value 2002
Consumer 1: Consumed value 2002 from Producer 2
Producer 1: Produced value 1002
Consumer 1: Consumed value 1002 from Producer 1
Producer 2: Produced value 2003
Consumer 1: Consumed value 2003 from Producer 2
Producer 1: Produced value 1003
Consumer 2: Consumed value 1003 from Producer 1
Producer 1: Produced value 1004
Consumer 2: Consumed value 1004 from Producer 1
Producer 2: Produced value 2004
Consumer 1: Consumed value 2004 from Producer 2
Producer 1: Produced value 1005
Consumer 2: Consumed value 1005 from Producer 1
Producer 2: Produced value 2005
Consumer 1: Consumed value 2005 from Producer 2
Producer 1: Produced value 1006
Consumer 2: Consumed value 1006 from Producer 1
Producer 2: Produced value 2006
Consumer 1: Consumed value 2006 from Producer 2
Producer 1: Produced value 1007
Consumer 2: Consumed value 1007 from Producer 1
Producer 2: Produced value 2007
Consumer 1: Consumed value 2007 from Producer 2
Producer 2: Produced value 2008
Consumer 2: Consumed value 2008 from Producer 2
Producer 1: Produced value 1008
Producer 1: Produced value 1009
Consumer 1: Consumed value 1008 from Producer 1
Producer 1: Produced value 1010
Producer 2: Produced value 2009
Producer 1: Produced value 1011
Consumer 1: Consumed value 1009 from Producer 1
Consumer 1: Consumed value 1010 from Producer 1
Consumer 2: Consumed value 2009 from Producer 2
Consumer 1: Consumed value 1011 from Producer 1
Producer 2: Produced value 2010
Producer 1: Produced value 1012
Consumer 2: Consumed value 2010 from Producer 2
Producer 2: Produced value 2011
Producer 2: Produced value 2012
Producer 2: Produced value 2013
Consumer 2: Consumed value 1012 from Producer 1
Consumer 1: Consumed value 2011 from Producer 2
Producer 1: Produced value 1013
Consumer 2: Consumed value 2012 from Producer 2
Consumer 2: Consumed value 2013 from Producer 2
Producer 2: Produced value 2014
Producer 1: Produced value 1014
Consumer 1: Consumed value 1013 from Producer 1
Consumer 1: Finished consuming 15 items
Consumer 2: Consumed value 2014 from Producer 2
Consumer 2: Consumed value 1014 from Producer 1
Producer 2: Finished producing 15 items
Producer 1: Finished producing 15 items
Consumer 2: Finished consuming 15 items
[1]   Done                    ./producer 1 15
[2]-  Done                    ./producer 2 15
[3]+  Done                    ./consumer 1 15
cconnell@cconnell:~/cs3502/assignment2/c-programs$
```

Reflection Questions

2.1

If you only use a mutex, there is nothing keeping track of the number of items in the buffer. The producer and consumer don't really know the state of the buffer. They just know if they get in there, they need to do what they were programmed to do. If the buffer is full and the producer enters the mutex it will cause an overflow when it tries to add to the full buffer.

2.2

If you acquire the mutex before waiting on empty/full you could be stuck holding the mutex and waiting to acquire empty/full that is never going to unlock. Say a producer fills the buffer then releases the mutex. Another producer can grab the mutex and will wait to acquire empty. Since no other processes can get the mutex while the producer is holding it, consumers cannot enter and the producer is stuck waiting on empty forever.

2.3

Since shared memory is accessing the same memory space it doesn't have to copy and move the data around. So, shared memory is faster than pipes. Shared memory is more complex to set up initially, but once it's set up, I think it would be more efficient, again since it's not having to move data around everywhere. I think shared memory would scale better than pipes. If you were setting up a lot of programs that need to communicate and they aren't parent/child processes, then I think in the end pipes would grow more complex than shared memory.

Pipes seem easier to set up, at least between parent and child processes. Especially since you don't have to worry as much about race conditions. They also seem less likely to cause deadlocks. Maybe because outside of closing the unused pipe ends the system handles a lot behind the scenes.

2.4

Video streaming sites like YouTube or Netflix and audio streaming sites use the producer-consumer pattern. When you click on a YouTube video, the server starts (among other

things) sending small chunks of video to a buffer. Your computer or media player retrieves the video from the buffer to play it. The player downloads from the buffer a little ahead of where you're at in the video (the little gray bar in front of the red on YouTube) to account for speed differences or quick interruptions. I guess that is why you when your video is lagging and loading, we say it's "buffering."

I think downloading a file would use the producer-consumer pattern too. It's not just saving on your computer, the producer is putting pieces in a buffer, and the consumer is picking it up and writing a file somewhere on your computer. Since this assignment is talking about concurrency and synchronization, I think download managers just increase the concurrency, just more connections downloading pieces of a file, to quicken the download. Each individual download is the same speed, but they use multiple ones to decrease the time it takes to download.